Figure 7.3 Use of web3 API

To apply and illustrate the web3 concept, sections 7.2 and 7.3 describe a channel concept and a micropayment channel Dapp for facilitating global plastics cleanup, and provide an end-to-end implementation. The micropayment channel is based on the example provided in the Solidity documentation. The planetary-level plastics cleanup application was created especially for this chapter.

7.2 The channel concept

You must recognize by now that blockchain is not intended to replace existing applications, but to address issues that are not solvable with traditional approaches. Don’t think of replacing your current system with a blockchain system or porting an existing application in one language (such as Java or Python) to Solidity simply to use the capabilities of blockchain technology. You have to be thinking about newer and never-before-seen application models for blockchain. Similar to the Dapps you learned about in chapters 4-6--for voting (ballot), holding an auction (blind auction), and trading in the marketplace (ASK airlines)--the channel concept you’ll learn in this chapter is ideally suited for blockchain-based decentralized applications. In section 7.3, you create a micropayment channel that enables new business models and encourages the participation of diverse people in an ecosystem emerging from blockchain technology.

The channel concept is ubiquitous in many domains, from geology to business. *A channel* is a path along which information passes from one point to another. In this chapter, you’ll use it for a payment mechanism. This concept is popularly known as the